pfi

Python for Fixed Income

COURSE 9 · DEPLOYMENT

Case Study: The Desk on a Hugging Face Space

WEEK 2 · DAY 5

Created by **Jeff Lenamon**

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Today: the desk on a Space

Today

1. Rebuild the snapshot, as day 9 did
2. What a Space is, and what is free today
3. space_folder: six files, built by code
4. check_space_folder: five planted problems
5. Case study: prepare, deploy, build log
6. Requests, the stage, and the delete
7. What the deployment shows, and where it fails

Recorded once, shown here

The notebook sends nothing to Hugging Face. The cloud steps ran once on the course's own free account, masked. You run them on yours, once your account is 30 days old.

A Space is a repository that also runs an app: the Hub builds it on every commit.

The rules for a public page

Every page states its date	figures as of 2026-10-02, with the disclaimer
Recorded questions only	anything else: not in the saved runs; a view is refused first
Synthetic holdings only	a real company only as the author of a document
No key on the Space	the app reads one JSON file; the token stays in .env

What the page shows

Allowed names

the brief as of 2026-10-02

the check table

the saved run

Never called

the desk's view

advice or a call

a recommendation

What a Space is, and what is free today

21	53	2	3
requests replayed	checks passed	checks failed	recorded answers kept

A Git repository on the Hub, with an app in it

Push a commit: the Hub builds the app and runs it. The README's YAML header says how.

ZeroGPU: runs on a CPU, borrows a GPU only inside @spaces.GPU. The desk's app never does.

Spaces: Gradio needs a paid plan; a free account may host 2 on ZeroGPU (huggingface.co/docs/hub/spaces-overview, read 2026-10-12).

ZeroGPU: an account older than 30 days; Python 3.12.12 or 3.10.13 (huggingface.co/docs/hub/spaces-zerogpu, read 2026-10-12).

ZeroGPU: 5 minutes of GPU a day on a free account, 40 on PRO (huggingface.co/docs/hub/spaces-zerogpu, read 2026-10-12).

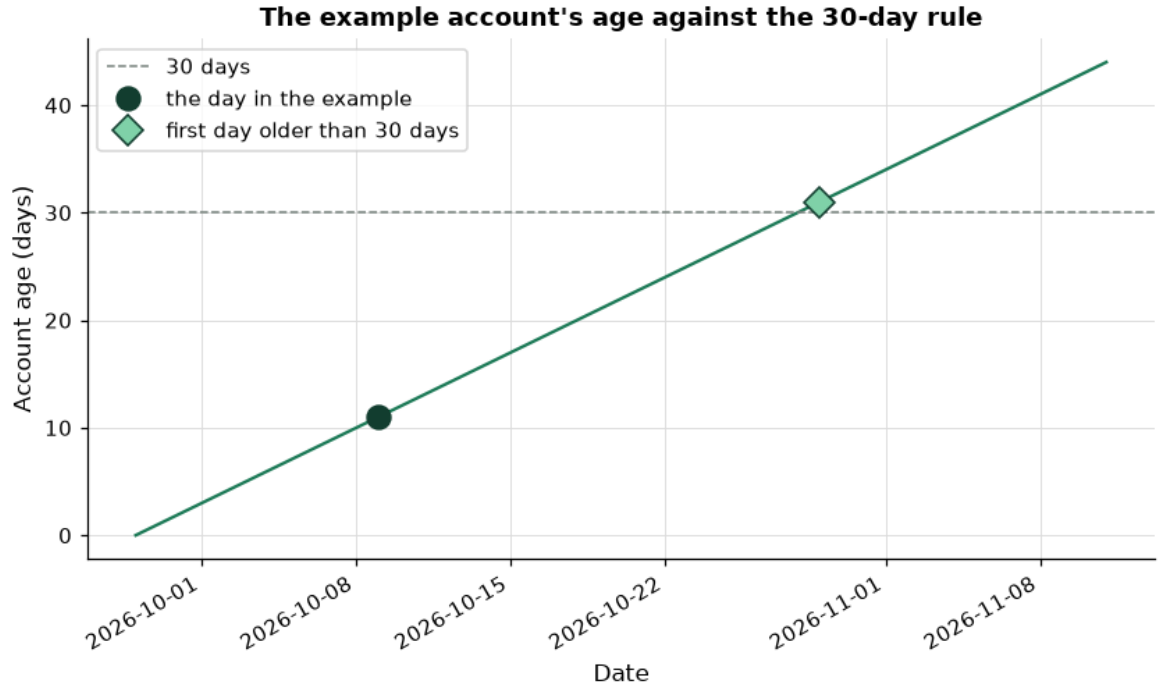
Hugging Face PRO: US$9 a month, hosts ZeroGPU, Gradio, and Docker Spaces (huggingface.co/pricing, read 2026-10-12).

```
account made 2026-09-28; older than 30 days from 2026-10-29; on 2026-10-09: not yet, 20 days to go
```

How a Space works

The problem	An app on your laptop reaches nobody else; a Space hosts it at an address a colleague can open
The method	A Git repository on the Hub: push a commit, the Hub reads the README's header, builds an image, starts app_file
ZeroGPU	The app runs on a CPU and borrows a GPU only inside @spaces.GPU, which the desk's app never does
You control	The header (sdk_version, python_version, app_file), the hardware, and the visibility
What comes out	A page at huggingface.co/spaces/<user>/<name>, a build log, and a stage a script can read

The account's age against the 30-day rule



Milestone	Date	Age	Over 30
account made	2026-09-28	0	no
the example day	2026-10-09	11	no
first over 30	2026-10-29	31	yes

Waiting is the only way to move along the line: make the account on day 1.

How space_folder works

The problem	The Hub needs exactly the files that start the app; a folder written by hand drifts from the tested module
The method	Writes app.py (placeholder, snapshot, gradio_app, launch), README.md (header and disclaimer), requirements.txt, the snapshot
The modules	Copies deploykit.py and bondmath.py beside them, unchanged: the Space runs what passed the tests here
You control	The folder and the snapshot
What comes out	The six files written, as paths, ready for the check and the upload

space_folder: six files, built by code

```
README.md 17 lines
app.py 18 lines
bondmath.py 572 lines
deploykit.py 1,084 lines
desk_snapshot.json 807 lines
requirements.txt 1 lines
```

Versions in quotes

```
python_version: 3.10 reads as 3.1
```

```
title: PFI Desk Brief
colorFrom: green
colorTo: blue
sdk: gradio
sdk_version: "6.30.0"
python_version: "3.12.12"
app_file: app.py
pinned: false
short_description: A descriptive desk brief from save...
```

No emoji key: the Space built and ran without it. requirements.txt: one comment, no gradio.

How check_space_folder works

The problem	An upload is public and a build takes minutes: catch what breaks a build or leaks a key before anything leaves
The method	Each version in the header is text; no gradio line in requirements.txt; the placeholder in app.py
Then day 7's rules	An environment file at any depth, a key-like string, an unpinned requirement, a folder over the size limit
Each finding	One sentence naming the file and the rule, never a value
What comes out	A list of sentences, empty when the folder is safe to upload; space_status: one line, stage and hardware

check_space_folder: five problems from four plants

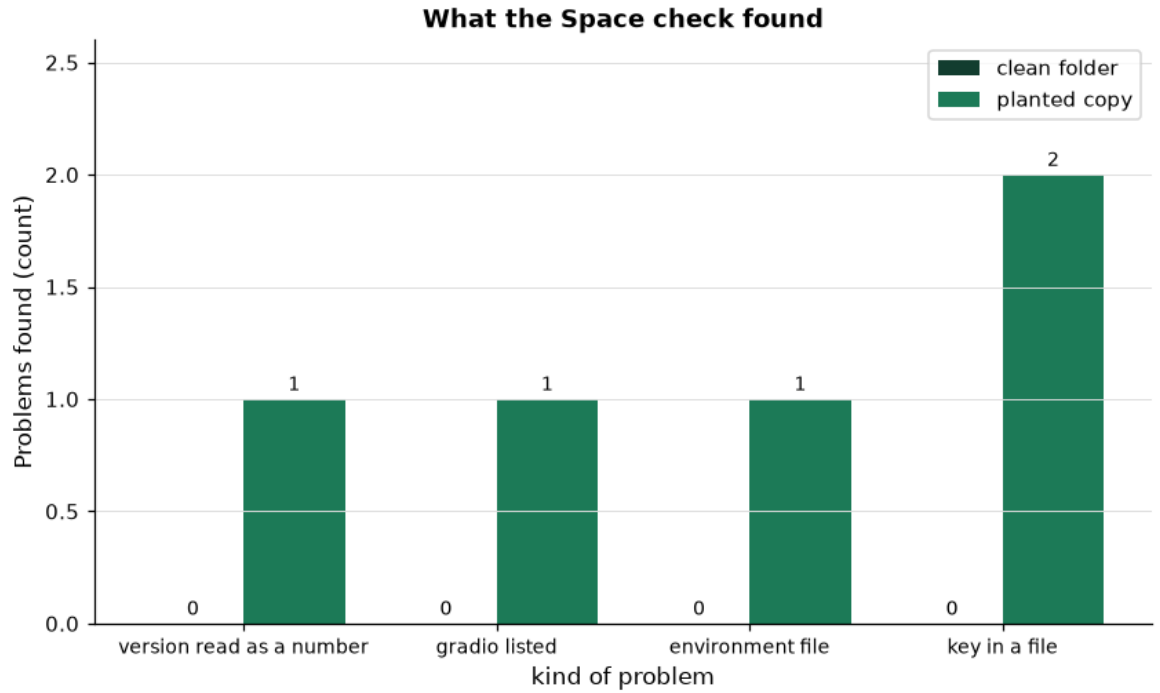
```
README.md: python_version 3.1 is not text; write it in quotes
requirements.txt: gradio is listed; the Space installs it from sdk_version
config/.env: an environment file at any depth is never deployed
config/.env, line 1: a key-like string (sk-or- pattern, starts 'sk-', 18 characters)
settings.py, line 1: a key-like string (hf_ pattern, starts 'hf_', 27 characters)
problems: 5; safe to upload: no
```

space_status: one line from get_space_runtime

```
Space stage RUNNING on zero-a10g.   Space stage SLEEPING on zero-a10g.
```

46 passed. A token shows its kind, three characters, and its length: never its value.

What the Space check found



Kind of problem	Clean	Planted
version read as a number	0	1
gradio listed	0	1
environment file	0	1
key in a file	0	2

Five problems from four plants, each found before anything is uploaded.

The case study: context, objective, data

The desk head: the brief on a page a colleague can open from anywhere, with its date, its checks, and a chat that refuses a view. No key.

```
file  bytes       sha256
         README.md    608 58718bafc7fa
            app.py    487 5c9bde0f1b75
       bondmath.py  30344 1f8940528441
      deploykit.py  59234 b500df21184b
desk_snapshot.json  25854 56f49195ab86
  requirements.txt     73 dafb73f4c91a
total 116,600 bytes; the snapshot hashes 21 saved exchanges
```

Preparation

```
public_scan rows  0

pins_check problems  0

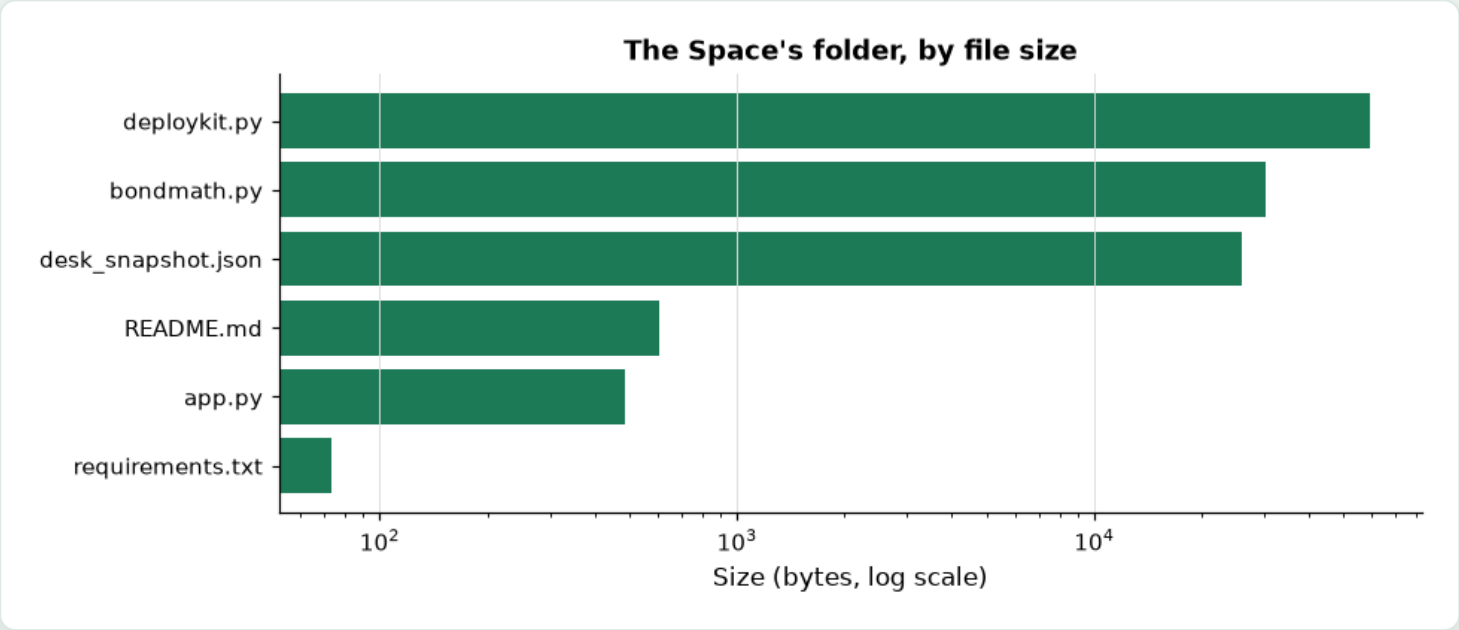
check_space_folder problems  0

README without the disclaimer  0

ready to upload: yes
```

Not uploaded: .env, the token, Course 8's saved runs, the desk's modules.

The six files, by size



File	Bytes
README.md	608
app.py	487
bondmath.py	30,344
deploykit.py	59,234
desk_snapshot.json	25,854
requirements.txt	73

The modules and the snapshot are the weight; the Hub reads three small files to start the Space.

How the deploy works

The problem	The folder has to reach the Hub, and you have to know when the app runs
Three steps	create_repo makes the Space (kept if it exists); upload_folder sends the six files as one commit; the Hub builds
The wait	get_space_runtime every 10 seconds, each change printed, until the stage leaves the waiting ones
You control	The hardware, the visibility, and exist_ok
What comes out	The Space's page, the commit's address, and the stages, ending in RUNNING or an error

The deploy: deploy_space.py

```
create_repo(space_id, repo_type="space",
    space_sdk="gradio",
    space_hardware="zero-a10g", exist_ok=True)
upload_folder(..., folder_path=FOLDER)
get_space_runtime(space_id)  # every 10 s
```

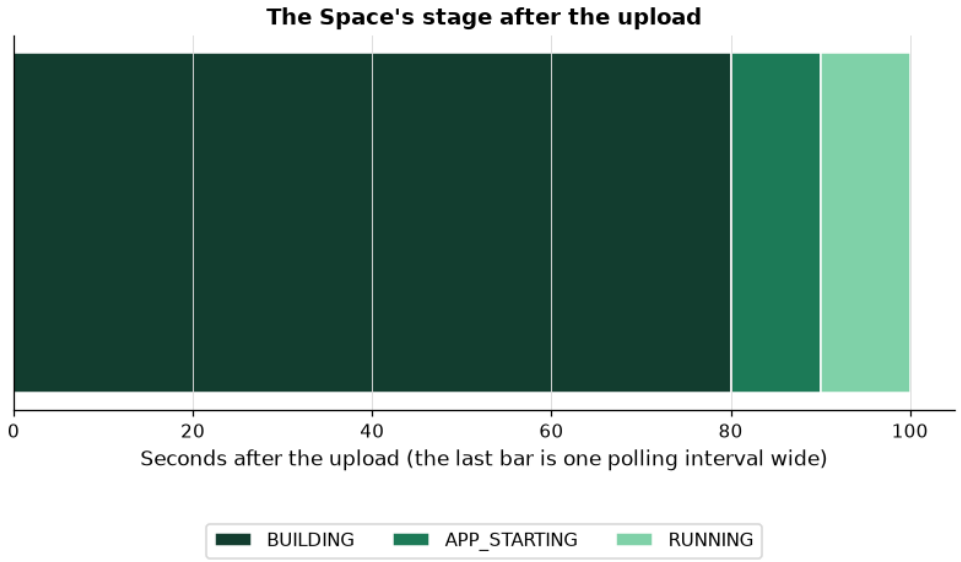
The probe: private on a free account?

```
created, private: https://huggingface.co/spaces/s...
```

Allowed. So the course's Space stayed private; yours is created public.

```
Space: https://huggingface.co/spaces/student/pfi-desk-brief
uploaded 6 files in one commit: https://huggingface.co/spaces/student/pfi-desk-brief/commit/c9969...
poll 1: stage BUILDING, hardware None, requested zero-a10g
poll 9: stage APP_STARTING, hardware None, requested zero-a10g
poll 10: stage RUNNING, hardware zero-a10g, requested zero-a10g
final stage: RUNNING
```

The stage after the upload



Stage	First poll	Seconds
BUILDING	1	0
APP_STARTING	9	80
RUNNING	10	90

Polled every 10 seconds; the first RUNNING poll was 90 seconds after the upload.

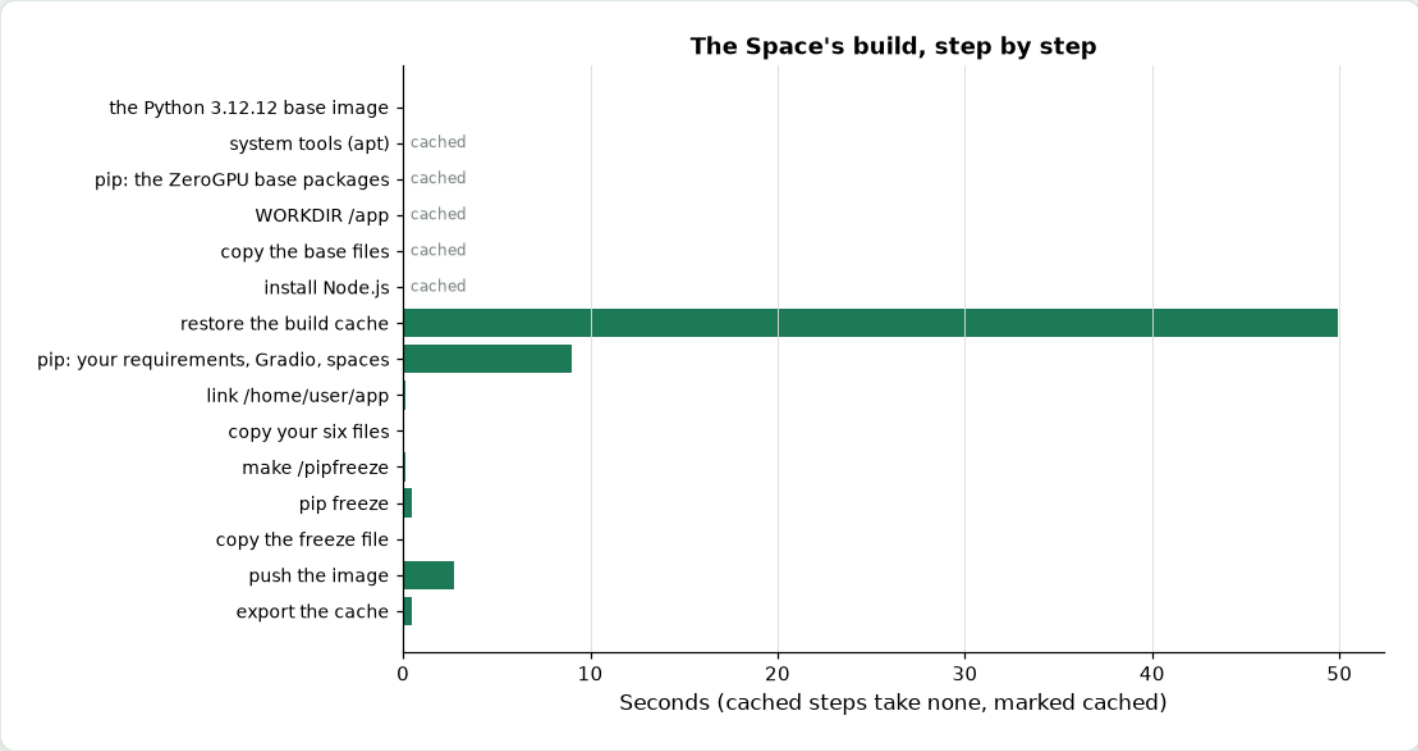
The build log

```
build log: 213 lines; the first 25 and the last 40
--> FROM docker.io/library/python:3.12.12@sha256:cb770de9f47e77715f13434d96f7ebaae7ad3a1f4fd9c8b3...
--> COPY --link ./ /app
--> Pushing image
```

```
pip install -r /tmp/requirements.txt "torch<=2.13.0" gradio[oauth,mcp]==6.30.0 "uvicorn>=0.14.0"
"websockets>=10.4" spaces==0.51.3
```

Package	The Space installed	The course pins
gradio	6.30.0	6.30.0
fastapi	0.143.0	0.143.0
pydantic	2.12.5	2.13.5

Where the build's time went



Longest steps	Seconds
restore the build cache	49.9
pip: your requirements, Gr	9.0
push the image	2.7
pip freeze	0.5
export the cache	0.5

One build, one day: the cache restore is the longest step, and cached steps take no time.

Three requests, the stage, and the delete

```
> python space_request.py
bond calculator, the week's bond:
Clean price: 99.737845 per 100 of face
Accrued interest: 0.000000 per 100 of face
DV01: 0.043625 price points per 100 of face
chat, a recorded question:
The 10 Yr Treasury par yield increased by 27 basis points between the date of the last FOMC post-...
(As of 2026-10-02 in the course's saved runs.)
chat, a request for a view:
REFUSED: this course asks only what the document says
```

The stage, then after about 20 minutes

```
Space stage RUNNING on zero-a10g.

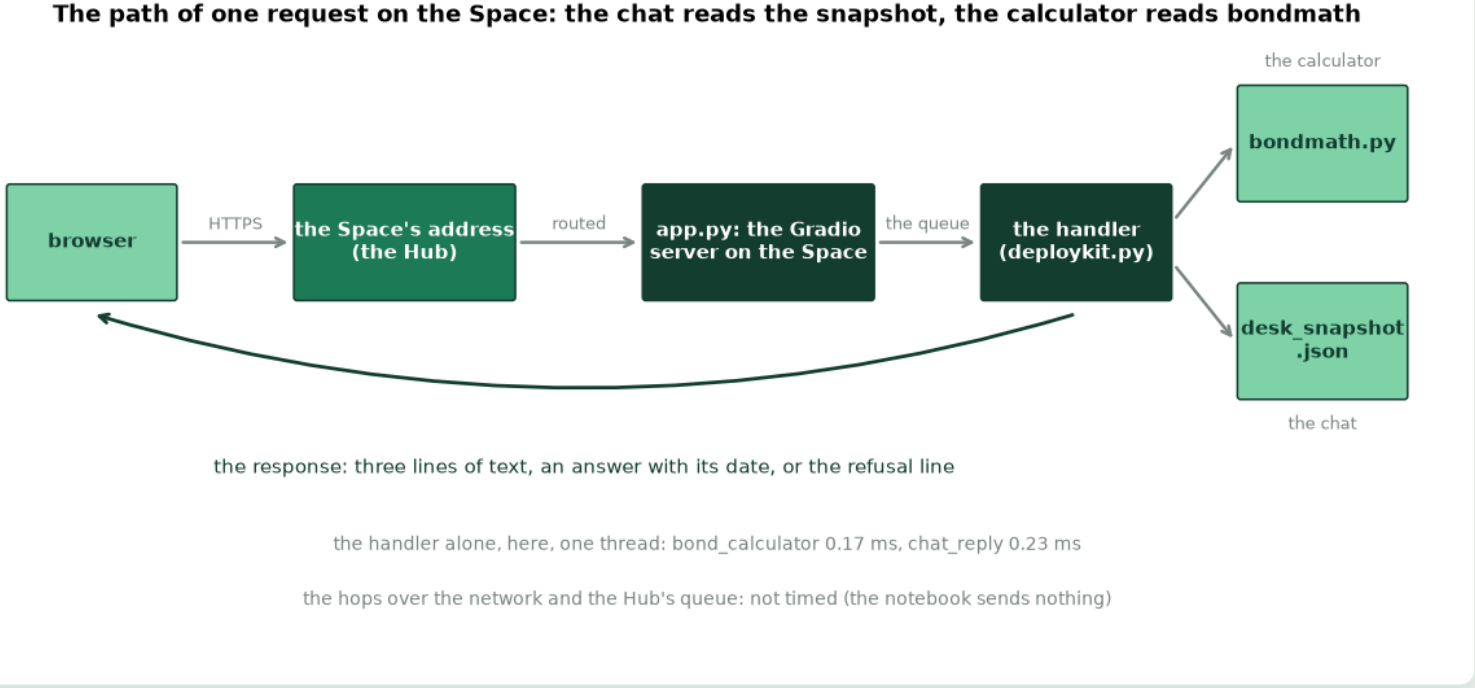
Space stage RUNNING on zero-a10g.
```

delete_space.py

```
deleted: student/pfi-desk-brief

the Space exists after the delete: no
```

The path of one request on the Space



Stop	Where it runs
browser	your machine
the Space's address	the Hub
app.py, the Gradio server	the Space (Python 3.12.12)
the handler	deploykit.py on the Space
the file	the uploaded files
the response	back the same way

The handler is where the requests part: the calculator reads bondmath.py, the chat reads the snapshot. Only the handler is timed, here, and drawn.

What the deployment shows, and where it fails

Shows: the brief, the checks, a calculator, and a chat, on free hardware, from six checked files, with no key; the same text there as here.

- Sleeps when unused; wakes slowly
- Disk is not persistent
- Every visitor shares one process
- Public code, unless private
- Python 3.12.12, not 3.13.8
- Two such Spaces, after 30 days
- Figures fixed at 2026-10-02
- Answers only what was recorded

```
calculator, three lines  same: yes
recorded question, its answer and date  same: yes
request for a view, the refusal line  same: yes
```

Spaces on cpu-basic sleep after 48 hours unused; a visit restarts them (huggingface.co/docs/hub/spaces-gpus, read 2026-10-12).

Spaces: 16 GB of RAM, 2 CPU cores, 50 GB of disk that is not persistent (huggingface.co/docs/hub/spaces-overview, read 2026-10-12).

Excel side by side, and the deploy log

Excel side by side (tested in Excel, on the course's private Space)

```
=WEBSERVICE("https://student.hf.space/gradio_api/info")
```

#VALUE! in Excel; HTTP 404 for Python with no sign-in. A public Space: not tested.

```
date target url_masked commit tests_passed bundle_mb status
2026-10-12 hugging face space, zerogpu https://huggingface.co/spaces/student/pfi-desk-brief c9969ca 46 0.11...
2026-10-12 hugging face space, zerogpu https://huggingface.co/spaces/student/pfi-desk-brief c9969ca 46 0.11...
read back from deploy_log.xlsx equal: yes
```

```
(HEAD -> main, tag: week2) The Space scripts and the deploy log
deploykit day 10: space_folder, check_space_folder, space_status, 46 tests
The earlier modules, frozen, and the ignore list
```

Tag week2: the end of the serving week.

The AI check

The docstring is the prompt. The deploy code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a requirements writer that pins gradio==6.29.1 "to be safe": a second Gradio on the Space.

Code that runs is not the same as code that is right.

What you can do now

- What a Space is; two ZeroGPU Spaces after 30 days
- A Space folder built by code, versions in quotes
- Checked before it leaves: gradio, YAML, keys
- Deploy, build log, requests, stage, delete
- Where it fails: sleep, disk, one process, public code

Week 3, Day 1 Docker: one image everywhere